

Utility Elicitation During Negotiation with Practical Elicitation Strategies

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Abstract—Automatic negotiation is gaining more interest recently thanks to the wider deployment of intelligent systems and the need for them to cooperate/compete on behalf of their users. A central assumption of most autonomous negotiation agents is that the utility function of the user is perfectly known to the agent. That is an often unmet assumption in real situations. Utility elicitation is the process of learning about the utility function of the user incrementally and has a long history in decision support research. Recently, some utility elicitation systems capable of incrementally eliciting the utility function of the user during the negotiation were presented. This work expands this body of research by optimizing the elicitation algorithm to realistic elicitation strategies. The proposed method extends the optimal elicitation algorithm to the – practical – case where queries to the user only reduce the uncertainty in the utility function without removing it completely. Extensive evaluation shows that the proposed extension outperforms two state-of-the-art elicitation algorithms and several baseline alternatives.

I. INTRODUCTION

Automatic negotiation is attracting more attention from the research community in recent years specially given the rise of AI, machine learning systems and the Internet of Things (IoT) that promise to automate most repetitive aspects of our lives. The main advantage of automatic negotiation in this context is providing the means for multiple actors to organize their behavior to achieve win-win deals allowing for better use of resources and providing an advantage for the society as a whole. It is becoming of enough interest to the research community that major conferences like AAMAS and IJCAI provide competitions in automatic negotiation (e.g. the ANAC competition [1]).

Recent applications of automatic negotiation include permission management in IoT systems [2], Wi-Fi channel assignment [3], providing feedback for student negotiation skills [4], and agriculture supply chain support [5].

Most of this work assumes that the negotiation agent has perfect knowledge of the utility function of the person/entity it is representing during the negotiation. While this can be the case in some limited cases, in many real-world scenarios; it is not possible to have perfect apriori knowledge of this utility function. This can be the case when the agent is representing a human subject or when the utility of a given outcome is difficult to calculate. Utility elicitation from people always involves some elicitation *cost* (bother cost). Even when asking

about the user utility is entailing a small cost per query, the user may not be able to provide an accurate estimate of her utility value for an outcome except using several queries which easily accumulates leading to a large elicitation cost.

Utility elicitation was studied extensively in the decision support community [6]–[8]. Most of this work focuses on the problem of eliciting the utility function of an actor for several possible outcomes of the negotiation process. Negotiation adds a new complexity to this problem because during negotiation it is not enough to know the utility value of some outcome for the user in order to propose/accept it but it is also essential to judge the probability that this outcome/offer is also acceptable by the partner(s) in the negotiation.

More recently, few works have reported systems for utility elicitation *during* negotiation [9], [10]. These systems build upon earlier work in *preference elicitation* in the decision support domain [11]–[13] while taking into account the specific features of negotiation.

The *optimal elicitation algorithm* [9] assumes that the actor (user) can be queried to provide *exact* utility values for different outcomes. This can be achieved using several possible elicitation strategies that does not require actual assignment of a numeric value to any outcome by the user so it is not restrictive. Nevertheless, this form of *deep* elicitation for each outcome considered is time-consuming and would lead to high levels of elicitation bother to the user that can sometimes be avoided. For example, if a single query assigned a probability distribution to an outcome that made it *dominated* by another there is no need to keep eliciting this dominated outcome until a numeric utility assignment is achieved. One advantage of this approach though, is the reliance on an easy to calculate *aspiration level* for the judgment of the expected utility for outcomes rejected by the partner(s). It is also efficient enough to be usable when time-pressure is an issue for the negotiation.

The *value of information elicitation algorithm* [10] avoids the problem of deep elicitation by assuming that a predefined set of possible queries are available that can reduce the uncertainty in the probability distribution of utility values for a given outcome. The system selects the optimal query at each point in terms of *expected expected utility* assuming a known acceptance model (opponent model). Because no *aspiration level* is utilized by this system, the probability of acceptance

of a given offer is calculated based on — an assumed to be accurate — opponent model. One disadvantage of this approach is that the agent must calculate a strategy of multiple consecutive outcomes to evaluate the probability of acceptance of any outcome. This increases the computational requirement of the algorithm significantly. Moreover, it is not trivial to decide the predefined set of queries.

In this paper, we extend this body of work by providing a middle ground between the optimal and value of information elicitation algorithms. The proposed system does not utilize deep elicitation (like the value of information elicitation algorithm), employs an aspiration level schedule to speedup the calculation of expected utility (like the optimal elicitation algorithm), and integrates well with known elicitation strategies from decision support research.

The proposed system is modular in the sense that different aspects of the interaction between negotiation and elicitation are separated allowing for easy experimentation of new variations for opponent modeling, elicitation strategies, stopping criteria, aspiration schedules, etc.

The rest of the paper is organized as follows: Section II provides more details about utility elicitation in the automatic negotiation context and provides the terminology used in the rest of the paper. Section III presents the proposed system which is evaluated in Section IV. The results of this evaluation are discussed in Section V and the paper is then concluded.

II. AUTOMATIC NEGOTIATION

A negotiation session is conducted between multiple agents representing self-interested actors over a set of issues. Issues can have discrete or continuous values. Every possible assignment of a value to each issue is called an *outcome* and during negotiation it may also be called an *offer*. If an agreement is reached the agreed upon outcome is called a *contract*. Each actor (which can be a human but may be not) is assumed to be self-interested with some internal *utility function* that assigns a numeric value (assumed to be normalized to the range 0 to 1 in this work) to every possible outcome. The actor wishes to maximize the utility she receives from the negotiation through the behavior of its representing agent. Every actor also has a predefined *reserved value* that she receives if the negotiation was broken either due to timing out or explicitly by one of the agents.

Negotiation sessions are conducted in *rounds* in which different outcomes are offered/judged by the agents according to some negotiation *protocol*. Negotiation protocols can be moderated with a moderator-agent that have a special role (and may have special access to the utility functions of the agents under their control) or unmoderated with all agents having similar roles. Negotiation sessions can end in an agreement (contract) or be broken without agreement. To provide an incentive for the agents to concede from the outcome with maximum utility, a time-limit is usually applied to the negotiation and the session is broken (times-out) automatically if no agreement is reached within a predefined real-time or number of rounds limit.

More formally, a basic negotiation scenario is characterized by a tuple $(m, \Omega, R, \tilde{U}_i, \hat{U}_i^0 \forall 1 \leq i \leq n)$ where m is the number of actors/agents joining the negotiation, $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ is a list of n possible outcomes of the negotiation which may or may not be countable. For the rest of this work, we assume that n is finite. $R: i \rightarrow [0, 1]$ is a mapping from actor i to a reserved value $r_i \equiv R(i)$ which is the utility received by the actor in case no agreement was reached.

Each actor i has an exact utility function $\tilde{U}_i: \Omega \rightarrow [0, 1]$ is a mapping from the list of outcomes Ω to real numbers between zero and one inclusive. This *real* utility function is assumed not to be known to the agent.

Each agent i starts with some probability distribution $\hat{U}_i^0(\omega_j): [0, 1] \rightarrow [0, 1]$ which represents the probability that $\tilde{U}_i(\omega_j)$ equals u for $0 \leq u \leq 1$ (i.e. $\tilde{U}_{ij}^0(u) \equiv \text{Prob}(\tilde{U}_i(\omega_j) = u)$). We use $\hat{U}_{ij}^t$ as a shortcut notation for $\hat{U}_i^t(\omega_j)$.

We assume that $\hat{U}_{ij}^0(\tilde{U}_i(\omega_j)) > 0$ which means that the initial assumptions of the agent about the utility function of the actor are consistent with reality. For the rest of this paper we will identify agents/actors by the letter i ($1 \leq i \leq m$) and outcomes by the letter j ($1 \leq j \leq n$). Superscript t indicates the time-step if not otherwise indicated.

As an example, consider a researcher Alice who is going to visit a research institute in a foreign country for 8 weeks starting on the third week of the year. She is employing an agent to negotiate with a housing company for renting a studio to use during the visit. The issues are: rent (100\$ to 115\$/week), starting week (2, 3, 4) and ending week (10, 11, 12). In this case, $m = 2$, $n = 6 \times 3 \times 3 = 54$.

A. Negotiation Protocol

Negotiations are carried out between agents according to a protocol. Several negotiation protocols have been proposed over the years. They can either be mediated [14] or un-mediated [15]. This work utilizes a slightly modified version of the un-mediated alternating offers protocol [15] but is not limited to it.

This protocol works as follows: An ordering of the agents is defined which we assume — without loss of generality — that it is the same as the agent index i . The first agent starts the negotiation by *offering* an outcome ω^0 which is visible to all other agents.

According to the alternating offers protocol, the next agent (with index $c \bmod n + 1$ where c is the current agent index) either *accepts* the offer, ends the negotiation, or proposes a new offer $\omega^1 \neq \omega^0$. We also allow the agent to *pass* this round declaring neither acceptance nor rejection. This is useful if the agent expects more information about the offer being currently negotiated to be available.

This process is continued until one of the following stopping criteria is met:

- **Agreement:** An offer ω^t receives $n - 1$ acceptances from agents other than the one that proposed it. In this case,

this offer is declared as the agreed upon *contract* and each agent i receives a utility of $\bar{U}_i(\omega^t) - C_i$ where C_i is the total cost incurred by the agent during the negotiation.

- **Timeout:** A predefined number of offer exchanges/rounds (N) or a predefined number of seconds T has passed since the beginning of the negotiation. In this case, each agent i receives a utility of $r_i - C_i$ where C_i is the total cost incurred by the agent during the negotiation (if any).
- **Failure:** Some agent ends the negotiation when it has the chance to respond to some offer. This case is treated the same as a timeout.

B. Agreement Model

The negotiation process complicates utility elicitation by the fact that it is uncertain whether offering or accepting an offer at a specific point of the negotiation will lead to an agreement or not. For example, if the agent is to decide between two outcomes: ω_1 with a certain utility of 0.9 and ω_2 with a certain utility of 0.2 near the end of a negotiation with a reserved value of 0.1, the decision will depend on how likely is it that these two outcomes will lead to an agreement. Assuming that this is the last round of the negotiation, if the agent *knows* that the probability of ω_1, ω_2 being accepted are 0.01, 0.9 respectively, then the expected utility of offering ω_1 is $0.9 \times 0.01 + 0.1 \times 0.99 = 0.108$ while for ω_2 it is $0.2 \times 0.9 + 0.1 \times 0.1 = 0.19$ and it should offer ω_2 as long as $r_i < 0.19$.

When only probabilistic information is available about the utility function then another level of expectation is needed and it can be argued that the most *rational* behavior in such cases — in a wide range of circumstances — is to use *expected expected utility* (EEU) [16].

EEU depends in general on two factors, the distributions $\hat{U}_{ij}$ and the acceptance probability of different outcomes. We formalize the later using the *agreement model* $\Lambda_i^t : \Omega \rightarrow [0, 1]$ which maps outcomes to the probability of them being accepted as contracts at the current time-step.

The agreement model Λ_i^t changes during the negotiation to reflect the exchanges between agents. In this work, we consider a very simple adaptive form of the agreement model that sets $\Lambda_i^t(\omega)$ to zero if ω was offered and rejected and increases $\Lambda_i^t(\omega)$ by $(1 - \Lambda_i^0(\omega))/(m - 1)$ whenever it is offered or accepted by any agent except i . In the special case of bilateral negotiation, this model collapses to the opponent model proposed in [9].

C. Expected Utility of an Offer

The expected expected utility of an offer for agent i at time-step t can be estimated as:

$$EEU_{ij}^t = \Lambda_i^t(\omega_j) \times \mathcal{E}(\hat{U}_{ij}^t) + (1 - \Lambda_i^t(\omega_j)) \times g_i^t(\omega_j), \quad (1)$$

where $\mathcal{E}$ is the *estimator* function which maps the probability distribution over utility values for some outcome to an estimate to the utility value to be received if that outcome becomes the contract, and $g_i^t(\omega_j)$ (estimate of utility on

rejection) is the current estimation of the utility to be received if it was not agreed upon.

In this work, we consider two ways to calculate the estimate of utility on rejection: aspiration and exact methods.

In the aspiration approach (used in [9]), $g_i^t(\omega_j)$ is estimated by a constant for all outcomes that depend only on the time of the negotiation. It starts with a large value and is gradually reduced toward the reservation value. A common method for calculating the aspiration value [9] is:

$$g_j^t = r_i + (g^0 - r_i) \times \left(1 - \frac{t^{1/e}}{N}\right), \quad (2)$$

where e is a parameter that controls the shape of the aspiration curve, g^0 is the maximum aspiration (at time step 0) and N is the maximum allowed number of steps. For $e > 1$, the aspiration level keeps near g^0 for a long time (*boulware*), for $e < 1$, it dips fast toward the reserved value (*conceder*) and for $e = 1$ it goes linearly from g^0 to r_i .

In the exact method (see [10]), $g_i^t(\omega_j)$ can be calculated as follows: firstly, the best possible ordering of outcomes (policy) π_j that does not contain the outcome ω_j is calculated as:

$$\pi_j = \arg \max_{\pi} \sum_{\omega \in \Omega/\omega_j} \mathbf{E}(\hat{U}_i(\omega)) \Lambda_{iK_{\pi}^{\omega}}^t \prod_{k=1}^{K_{\pi}^{\omega}-1} (1 - \Lambda_{i\pi(k)}^t), \quad (3)$$

where K_{π}^{ω} is the index of outcome ω in the policy π that does not contain ω_j , $\pi(k)$ is the outcome with index k in the policy π , and $\mathbf{E}$ is the expectation operator (mean).

Given π_j , it is possible to estimate $g_i^t(\omega_j)$ as:

$$g_j^t(\omega_j|\pi_j) = \sum_{\omega \in \pi_j} \mathbf{E}(\hat{U}_i(\omega)) \Lambda_{iK_{\pi_j}^{\omega}}^t \prod_{k=1}^{K_{\pi_j}^{\omega}-1} (1 - \Lambda_{i\pi_j(k)}^t) \quad (4)$$

D. Practical Elicitation Strategies

When the agent has full knowledge of its actor's real utility function, then no elicitation is required. In our terminology that represents the case when $\hat{U}_{ij}^0 = \delta[u = \bar{U}_i(\omega_j)] \forall 1 \leq j \leq m$ where $\delta[a = b]$ equals 1 whenever $a = b$ and zero otherwise.

When elicitation is required, there are several methods to query the actor about her utility function. These different methods are called *elicitation strategies* hereafter. Asking users to provide numeric utility values for specific outcomes (specially for multi-issue negotiations) is unreliable [8]. People are usually better at comparing different outcomes than providing exact utility values for them. Research in effective elicitation strategies goes back for decades and is carried out under names like search procedures [8], utility elicitation procedures [17], preference elicitation strategies, etc.

There are two general kinds of elicitation strategies in the literature: probability equivalence (PE) and certainty equivalence (CE) methods. Both rely on the notion of a *gamble*. A gamble is defined by a tuple (ω^*, ω_*, p) which means getting outcome ω^* with probability p and ω_* with probability $1 - p$.

In the special case when $p = 1$, the gamble is represented by its only possible outcome ω^* .

In both PE and CE methods the agent is asked to compare two gambles ω and (ω^*, ω_*, p) (i.e. getting ω with certainty or the gamble between ω_* and ω^*). The main difference between the two approaches is what the user is going to estimate.

For PE methods, the actor is asked to specify a value for p to make the two gambles equivalent given $\omega^*, \omega_*, \omega$ while for CE methods, she is asked to estimate a value for ω that makes the gambles equivalent given ω^*, ω_*, p .

Different strategies within the PE and CE frameworks differ in how they select these gambles. The most widely used PE method utilizes *extreme gambles* where ω^* and ω_* are selected to be the absolute and worst outcomes respectively and are assigned a utility of one and zero. These gambles are called sometimes *standard gambles* [12]. Other possibilities include *adjacent gambles* and assorted gambles [17]. CE methods like fractile and chaining methods differ in how they select the endpoints (ω^* and ω_*) as well as the probabilities used in the gambles. For a recent survey, please refer to [17]. Hershey et al. found a systematic difference between PE and CE methods (the response mode bias phenomenon) [18] and more recently, Schwand et al. found that CE and PE methods differ in how they are affected by changes in probability levels. For example, if the problem is formulated in the loss domain, then PE indicates a more risk averse behavior with increasing probability levels while CE shows the converse effect (the tailwhip effect) [19]. For a recent discussion of these effects and related biases (e.g. the range bias), refer to [20].

Indicating the probability p or the certain outcome ω in PE and CE methods is still a challenge for most naive actors and these indifference points are usually elicited using multiple questions that each require the actor to indicate only which of two alternatives she prefers.

Three of the most widely used approaches here are *titration-up*, *titration-down* and *ping-pong* [8]. In these three approaches, the actor is asked to indicate which of two gambles she prefers: outcome ω with certainty or a gamble $(\omega^{best}, \omega_{worst}, p_k)$ with some schedule of p_k values. The point at which she changes her preference indicates the approximate utility value for outcome ω . For titration-up, $p_k = s \times k$ for a small value s (usually around 0.05). For titration down, $p_k = 1 - s \times k$, and for ping-pong $p_k = s \times \lfloor k/2 \rfloor$ for odd value of k and $p_k = 1 - s \times k/2$ for even values of k .

We model utility elicitation with a process Γ_i for actor i that receives a set of probability distributions over utility values for all outcomes $U_{ij}^t, \forall 1 \leq j \leq m$ and generates another set of probability distributions $U_{ij}^{t+1}, \forall 1 \leq j \leq m$ — according to some predefined rule — where the *uncertainty* of the utility value for at least one outcome j^t is reduced. This formulation can easily represent any of the elicitation strategies discussed above.

In most cases, the elicitation process Γ_i will modify the utility distribution for a single outcome j . All the elicitation procedures discussed earlier in this section are of this form.

In such cases, we represent the elicitation process as a functor that maps a utility distribution into another incurring some cost ($\hat{U}_{ij}^{t+1} = \Gamma_i[\hat{U}_{ij}^t]$). A deep elicitation process (Γ_i^D) is a special case in which the uncertainty in the utility value is removed (i.e. $\Gamma_i[\hat{U}_{ij}^t] \approx \delta[u = \tilde{U}_i(\omega_j)]$). A deep elicitation process is inherently slower and more costly than a shallow one as it requires the user to answer several questions.

E. What to elicit?

In Alice's example, we can easily see that ω^{best} is (110\$, 3, 11) because Alice will pay the minimum amount for exactly the time she needs. ω^{worst} will depend on the market and Alice's preferences. For example, if it is really hard to find a room for less than 115\$ starting from week 10 of the year, then the worst outcome will be (115\$, 1, 10).

Consider Alice's agent and assume that it is the first agent to propose an offer. Its current knowledge of Alice's utility function is encoded as:

$$\hat{U}_i^0(\omega) = \begin{cases} 1 & \omega = \omega^{best} \\ 0 & \omega = \omega^{worst} \\ U[0, 1] & \text{otherwise} \end{cases} \quad (5)$$

where $U(b, e)$ is a uniform distribution between b and e .

What are the questions that the agent can ask Alice in order to reduce this uncertainty? The elicitation strategy Γ_i will dictate the answer to this question. Assuming that we are using *titration-down* with step 0.05, then the immediate next set of questions are standard gambles between each outcome with uncertainty and the gamble $(\omega^{best}, \omega^{worst}, 0.95)$. This gives 52 possible questions.

Which of these questions should be asked first? This depends on the agreement model Λ_i^t . For example, if the agent knows — based on earlier encounters or market analysis — that fifty out of the 52 outcomes will be rejected and the agreement probabilities of the other two are the same, then it only makes sense to elicit one of these two outcomes and it does not matter which one.

This simplistic example, shows two things: the elicitation strategy determines the questions available to the agent and the optimal question to ask depends on both the current utility distributions over different outcomes as well as the agreement model.

III. PROPOSED ELICITATION ALGORITHM

During negotiation, the agent is expected to either *propose* outcomes to other agents or respond to their offers by acceptance, rejection, pass, or ending the negotiation (see section II-A). To simplify the presentation, we will focus on the case in the alternating offers protocol where only one outcome needs to be responded to/offered. Extension to protocols requiring the evaluation/proposal of multiple outcomes is trivial.

The proposed method is an extension of the *optimal elicitation algorithm* [9] to avoid deep elicitation (see section II-D) and take into account the elicitation strategy.

A. Optimal Elicitation Algorithm

The optimal elicitation algorithm [9] is based on Pandora's Rule [21] which solves the following problem: Given a set of n boxes with some probability distributions on the utility achievable from them $p_j : \mathcal{R} \rightarrow [0, 1]$ that will be received in the future with a discount factor $\beta_j \in [0, 1]$ and that can be opened with a cost c_j , find an optimal ordering of the boxes to open and decide when to stop opening boxes. Weitzman [21] proved that all the information about the box can be summarized with a single value $z_j \in \mathcal{R}$ called the *reservation price* which can be calculated by solving:

$$c_j = \beta_j \int_{z_j}^{\infty} (u - z_j) p_j(u) du - (1 - \beta_j) z_j \quad (6)$$

Pandora's Rule is to open the box with highest reservation price (z_i) if its reservation price is larger than the best utility for all open boxes and stop opening boxes if no such box can be found.

Baarslag and Gerding [9] adapted this rule to the elicitation problem by considering one box per outcome and defining the distributions p_j as follows:

$$p_j^t(u) = \Lambda_i^t(\omega_j) \times \hat{U}_{ij}^t + (1 - \Lambda_i^t(\omega_j)) \times g_i^t(\omega_j), \quad (7)$$

where g_j^t is defined using Equation 2. Opening a box was assumed to lead to the *exact* utility value for that outcome ($\hat{U}_{ij}^t \rightarrow \tilde{U}_i(\omega_j)$). Moreover, it is assumed that an open box always exists with no associated outcome and a utility value equals to r_j .

As the discussion in section II-D shows the exact utility assumption is not practical because a single question to the user does not usually lead to an exact utility value but is more likely to lead to a reduction in the amount of uncertainty in the utility distribution for some outcome. Under these circumstances, deep elicitation is not optimal as the new value of p_j after a single elicitation question (Equation 7) may lead to a change in the reservation price (according to Equation 6) that changes the optimal order and makes any more elicitation of this outcome unnecessary.

B. Proposed Extension

The simplest extension of the optimal elicitation algorithm (Section III-A) is to recalculate the reservation price (Equation 6) after every question is answered by the user and to use the expected value of the returned utility ($\mathbb{E}(\hat{U}_{ij}^{t+1})$) for the stopping decision.

The problem with this approach is that it does not take into account the time pressure. Early in the negotiation, the agent should try outcomes that have high *minimum* utility and near the end it should be more accepting of outcomes that have high *maximum* utility. Notice that this was not a problem in [9] because of the assumption that an exact utility function is returned from the user query needing no estimation of the final utility.

We propose to use Equation 1 for this purpose with the following heuristic estimator:

$$\mathcal{E}(\hat{U}_{ij}^t) = \frac{t}{N} \times \text{Min}(\hat{U}_{ij}^t) + \left(1 - \frac{t}{N}\right) \times \text{Max}(\hat{U}_{ij}^t) \quad (8)$$

where Min/Max is a *biased* estimator that exaggerate the lower/upper part of its input probability density function. For a uniform distribution $U(a, b)$, $\text{Min}(U) = a$, $\text{Max}(U) = b$. In the rest of this paper, we use uniform distributions and assume these forms but it is possible to extend this definition to other distributions.

To speedup the evaluation of reservation prices (Equation 6), it is possible to find a closed form solution for some probability distributions. As we assume a uniform distribution for $\hat{U}_{ij}^t$ throughout this work (the least informative), this solution can be found for $U(a, b)$ and cost c_j as:

$$z_j = \begin{cases} \frac{\frac{a+b}{2}\beta - c_j}{-\lambda \pm \sqrt{\lambda^2 - 4\zeta}} & z_j \leq a \\ a < z_j \leq b \end{cases}, \quad (9)$$

where $\lambda = -2 \left(b + \frac{a-\beta}{\beta}(b-a)\right)$ and $\zeta = b^2 - \frac{2c_j}{\beta}(b-a)$. Due to space limitations, the proof of this result will not be presented.

IV. EVALUATION

This section provides a set of experiments testing the proposed extension to the optimal elicitation algorithm against baseline and state-of-the-art approaches. In all cases, we employ a bilateral negotiation with an opponent that accepts only a predefined set of offers (unknown to the agent) and proposes randomly from them. All sessions used the alternating offers protocol and all elicitation algorithms for every experiment were tested using the same settings for everything except the elicitation algorithm itself.

Each scenario was generated by selecting a value for n , and N . For the eliciting agent, $\tilde{U}_j \sim U(a_j, b_j)$ where $a_j \sim U(0, 1)$ and $b_j \sim U(a_j, 1)$. For known distributions cases, $\hat{U}_{ij} = U(a_j, b_j)$ otherwise $\hat{U}_{ij} = U(0, 1)$. The acceptance list of opponents were selected from Ω where each outcome is included with probability p^a (set by the experimenter). For perfect opponent model cases, $\Lambda_i^0(\omega_j)$ was set to 1 for outcomes in the acceptance list and 0 otherwise. For other cases, $\Lambda_i^0(\omega_j)$ was set to 0.5 for all scenarios.

A scenario is then specified by the following parameters: Number of outcomes n , Maximum number of rounds N , Maximum real-time T , probability of outcome inclusion in the opponent acceptance list (p^a), whether or not the elicitation algorithm knows the distributions from which its actor's real utility was sampled and whether or not it has access to a perfect opponent model.

A. Algorithms

We selected three baseline algorithms: full knowledge, full elicitation, and random elicitation. The full knowledge agent needed no elicitation as it has access to the exact utility

function of the actor. The full elicitation agent, elicited all outcomes until it got approximately exact utility values for all of them before engaging in the negotiation. The random elicitation agent, elicited one outcome (randomly selected) whenever it is asked to offer an outcome. In all cases, the agent stopped eliciting and ended the negotiation once its expected utility after the elicitation (taking the cost into account) was less than the reserved value. Moreover, baseline algorithms always received the real utility value after deep elicitation.

Two state-of-the-art algorithms were used as benchmarks: The optimal elicitation algorithm with deep elicitation as described in section III-A and the Value of Information elicitation algorithm recently proposed in [10] which implicitly utilized the exact method for estimating utility on rejection (Equation 4) and does not utilize the aspiration level (Equation 2).

Other than the proposed balanced elicitation algorithm, we use the naive extension of the optimal elicitation algorithm using the mean of the utility distribution (Section III-B).

B. Comparative Study

In this study, we directly compare the performance of the proposed elicitation algorithm against the six algorithms described in section IV-A. The negotiation scenarios used either 10 or 100 outcomes (n) and allowed either 10 or 100 steps (N). A real-time limit of $600ms \times \max(n, N)$ was allowed. For each of these settings, p^a was varied between 0.1, 0.5, 0.9 to vary the probability of agreement. For each of these, the four conditions for knowledge of the actor's utility distribution and opponent acceptance model were tested. Elicitation cost was set to a constant value per query and varied between 0 and 0.25 with a step of 0.0125 for all algorithms except *VOI* for which the step was doubled. Three strategies were tested for each of these settings: *titration-down*, *titration-up* and *ping-pong* (see Section II-D). The aspiration type was set to *Boulware* for algorithms that uses it. Each of these scenarios was repeated 3 times because of the stochasticity of opponent behavior. This led to a total of 67,824 sessions.

Fig. 1 shows the results of the comparative analysis. As expected, the baseline algorithms (full knowledge, full, and random) achieve very high utility at zero cost because of their access to the real utility function of the user. The realistic alternatives (optimal, *VOI*, mean and balanced) cannot achieve this high performance due to the inevitable approximation in their estimates of the utility function.

The optimal and mean algorithms outperformed *VOI*, full and random algorithms for any nonzero elicitation cost. Fig. 1-A also shows that the proposed balanced algorithm is the best performing elicitation algorithm for any nonzero cost. Statistical analysis using factorial t-test with multi-comparison corrections show that this difference is statistically significant for nonzero costs up to 0.075 for all comparison algorithms and for all nonzero costs for the full, random and *VOI* elicitation algorithms.

Fig. 2 focuses on the evaluation of the proposed balanced algorithm in comparison with the two state-of-the-art algorithms (optimal [9] and *VOI* [10]). Fig. 2-A shows that the proposed

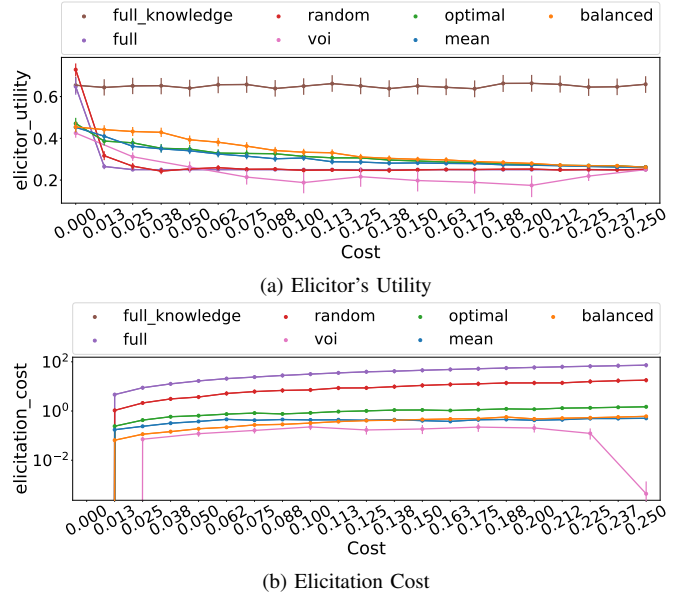


Fig. 1. The performance of the seven elicitation algorithms in the comparative study.

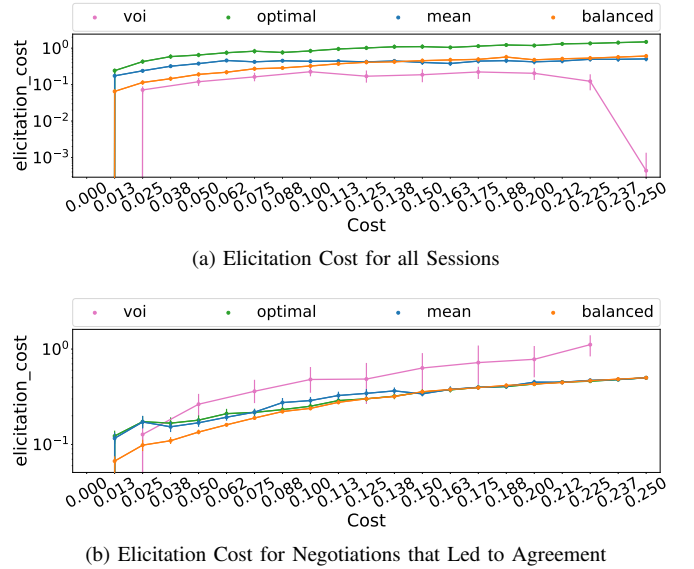


Fig. 2. The elicitation cost incurred by different algorithms.

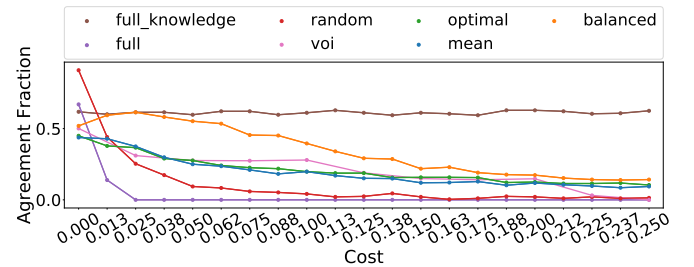


Fig. 3. Fraction of the sessions that ended in agreement.

method incurred a total cost that is in the middle between the *VOI* method which tended to elicit too little and the optimal

elicitation method which tended to elicit too much. Moreover, Fig. 2-B shows that the proposed algorithm actually incurred the minimum total elicitation cost for negotiations that led to agreement.

Fig. 3 shows the fraction of sessions that ended in agreement for different algorithms. The first finding is that for zero elicitation cost, the random algorithm achieves higher agreement than even the full knowledge and full elicitor. The reasons is that this algorithm with a single elicitation may offer the opponent suboptimal outcomes — in terms of the true utility function of the actor — which may lead to agreements that are not possible with full elicitation. More importantly, the proposed balanced algorithm is clearly the one with maximum agreement fraction of all elicitation algorithms (the difference is statistically significant for all nonzero costs under 0.175).

The balanced elicitation level and higher agreement potential, taken together, suggest that the proposed balanced elicitation algorithm can balance the need to query the user for more information about her utility with the need to reduce the elicitation cost more effectively than other tested approaches which explains its superior performance (Fig. 1-A).

C. Scalability Analysis

The goal of this experiment was to compare the scalability of performance for the proposed balanced elicitation algorithm and the state-of-art optimal elicitation algorithm [9]. The settings for this experiment was the same as for the comparative study but only the balanced, and optimal algorithms were tested. Moreover, the number of outcomes was varied between 10, 100, 1,000, 10,000 while the maximum number of rounds was varied between 10, 100, 1,000.

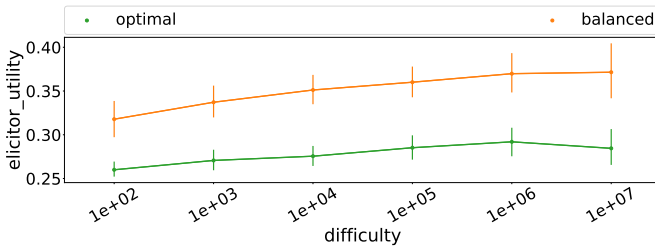


Fig. 4. Effect of increasing search space (e.g. number of outcomes or available negotiation time) on the eliciting agent's utility.

Fig. 4 shows the results of this scalability analysis. The improvement of the proposed technique is maintained (and slightly increased) with both increased number of outcomes and increased negotiation rounds allowed.

D. Effect of Scenario Parameters

1) *Elicitation Strategy*: Fig. 5 shows the difference in performance between the three tested elicitation strategies (ping-pong, titration up, and titration down). Titration down results in higher final utilities for small elicitation costs while ping-pong always results in the lowest final utility.

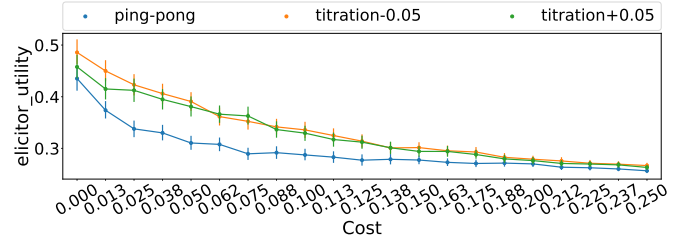


Fig. 5. Effect of Elicitation Strategy

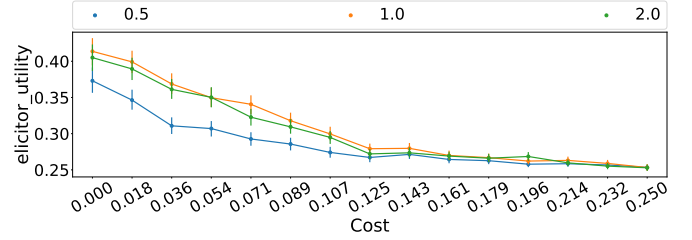


Fig. 6. Effect of Relative Maximum Aspiration (g^0/p^a)

2) *Aspiration Strategy*: Setting the maximum aspiration level at the beginning of the negotiation (g^0) is an important factor in both the optimal elicitation algorithm and the proposed balanced algorithm. Fig. 6 shows that setting it too low would result in lower final utility for the eliciting agent for mostly all costs. The best results can be achieved when setting $g^0 = p^a$ but setting g^0 to double this value did not affect the results much. Given that the willingness of the opponent to accept offers (p^a) is not usually available to the agent, these results suggest setting g^0 to a high value near 1.

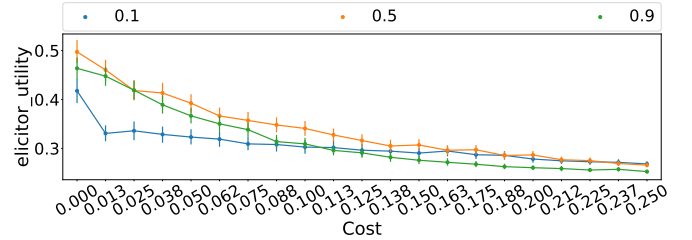


Fig. 7. Effect of the Opponent's Agreement Probability (p^a)

3) *Opponent's Agreement Probability*: It is interesting to study the effect of the opponent's willingness to agree (measured by p^a) on the final utility obtained by the eliciting agent. Fig. 7 shows – unsurprisingly – that agents that lowest utilities were obtained at low cost levels when the opponent tended to refuse most offers ($p^a = 0.1$) from the eliciting agent. Opponents that tended to agree most of the time ($p^a = 0.9$) resulted in lower utility in most cases than agents that accepted roughly half of the offers. This final effect may be the result of immature agreement before the eliciting agent is able to discover outcomes with guaranteed high utility.

4) *Reserved Value*: Increasing the reserved value is expected to increase the attainable utility of all elicitation al-

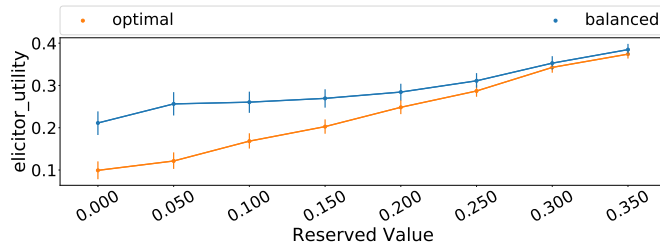


Fig. 8. Effect of the reserved value (r_i)

gorithms as negotiations that time-out or get broken receive higher values. Fig. 8 shows the effect of varying the reserved value between 0.0 and 0.35 for the optimal elicitation algorithm and the proposed balanced algorithm. The lower the reserved value, the more can the agent benefit from employing the proposed balanced elicitation algorithm.

V. DISCUSSION, LIMITATIONS AND FUTURE WORK

This paper presented an extension of the optimal elicitation algorithm to practical scenarios. Evaluation on a variety of scenarios with variable opponent agreement probability, knowledge of the agreement model and distributions of the user utility show that the proposed extension can more effectively balance the need to elicit more information about the utility function with the elicitation cost. This is achieved by eliminating the need for deep elicitation from the optimal elicitation algorithm without increasing the computational cost.

One limitation of all the algorithms discussed in this paper is the need to specify a utility distribution for every outcome which makes them applicable only to cases with countable outcome spaces. In many realistic negotiation scenarios, the outcome space is continuous. Moreover, negotiations involving multiple issues usually exhibit some structure in the utility function [14] that can be exploited during elicitation. In the future, extensions of the proposed system to multi-issue structured and continuous outcomes spaces will be explored. It is only interesting to quantify the effect of the level of uncertainty on the results of the negotiation as compared with full knowledge as well as human-human direct negotiations. Comparisons of different elicitation algorithms in real-world negotiation situations will also be considered.

VI. CONCLUSION

This paper presented an extension of the optimal elicitation algorithm based on Pandora's Rule to the practical case where queries to the user result in a reduction in the uncertainty of her utility for different outcome. The proposed method avoids deep elicitation and uses a heuristic approach to balance exploration of outcomes with higher guaranteed utilities with elicitation of more outcomes. Evaluation on a variety of generated scenarios show that the proposed method can result in higher agreement potential, lower elicitation cost in cases of agreement, balanced total elicitation cost and higher utilities compared with two state-of-the-art algorithms as well as a simpler variant and three baseline approaches.

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